Math 299 Notes

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1 Zorn's Lemma

1.1 Partially Ordered Sets

Definition (Partially Ordered Set, Chain)**.** A **partially ordered set** is a set M on which there is defined a **partial ordering**, that is, a binary relation which is written $\leq$ and satisfies the conditions

(PO1) $a \leq a$ for every $a \in M$ (Reflexivity)

(P02) If $a \leq b$ and $b \leq a$, then $a = b$. (Antisymmetry)

(PO3) If $a \leq b$ and $b \leq c$, then $a \leq c$ (Transitivity)

- The term "partially" means that there may exists elements a and b such that neither $a \leq b$ nor $b \leq a$. When this is the case, we call the set M to be **incomparable**.
- On the other hand, we say that a and b are **comparable** if they satisfy $a \leq b$ or $b \leq a$ (or both).

Definition (Totally Ordered Set/Chain). We call a set M to be **totally ordered** or **chain** if it is a partially ordered set such that every two elements of the set are comparable.

Definition (Upper Bound). An **upper bound** of a subset W of a partially ordered set M is an element $u \in M$ such that

$$x \leq u \;\; \forall x \in W.$$

- Another way to think about a chain is that it does not contain any elements that are incomparable.
- Note that the depending on the properties of M and W, the existence of such an element may or may not exist.

Definition (Maximal Element). We call a number $m \in M$ a **maximal element** of M if

$$m \leq x \implies m = x.$$

Similarly, M may or may not have maximal elements and that they need not be an upper bound.

Theorem (Zorn's Lemma). Let $M \neq \emptyset$ be a partially ordered set. Suppose that every chain $C \subseteq M$ has an upper bound. Then M has at least one maximal element.

The above is to be taken as an axiom.

1.2 Applications

Theorem (Existence of a Hamel Basis). Every vector space $X \neq \{0\}$ contains a Hamel basis.

Proof. Let M be the set of all linearly independent subsets of X. Since $X \neq \{0\}$, there exists an element $x \neq 0$ and $\{x\} \in M$ such that $M \neq \emptyset$. Set inclusion defines a partial ordering on M. Every chain $C \subseteq M$ has an upper bound, namely, the union of all subsets of X which are elements of C. By Zorn's Lemma, M contains a maximal element B.

Now, we will show that B is a Hamel Basis for X. Let $Y = \text{span}B$. Then Y is a subspace of X, and $Y = X$ since otherwise $B \cup \{z\}$ where $z \in X$ and $z \notin Y$, would be a linearly independent set containing B as a proper subset, contrary to the maximality of B. ∎

Before we go over the second example pertaining to Orthonormal sets, we recall some terms used within the proof.

Definition (Total Orthonormal Sets). A **total set** (or **fundamental set**) in a normed space X is a subset $M \subseteq X$ whose span is **dense** in X. Accordingly, an orthonormal set (or sequence or family) in an inner product space X which is total in X is called a **total orthonormal set** (or a sequence or family, respectively) in X.

Definition (Total Orthonormal Set). In every Hilbert space $H \neq \{0\}$, there exists a total orthonormal set.

Theorem (Totality (Theorem 3.6-2)). Let M be a subset of an inner product space X. Then:

(a) If M is total in X, then there does not exist a nonzero $x \in X$ which is orthogonal to every element of M; briefly,

$$x \perp M \implies x = 0.$$

(b) If X is complete, that condition is also sufficient for the totality of M in X.

Proof. Let M be the set of all orthonormal subsets of H. Since $H \neq \{0\}$, it contains an element $x \neq 0$, and an orthonormal subset of H is $\{y\}$, where $y = \|x\|^{-1}x$. Thus, $M \neq \emptyset$ and that set inclusion defines a partial ordering on M. Every chain $C \subseteq M$ has an upper bound, namely, the union of all subsets of X which are elements of C. By Zorn's Lemma, M contains a maximal element F.

We will show that F is total in H. Suppose for sake of contradiction that F is NOT total. Then using Theorem 3.6-2 (see book for details), there exists a nonzero $z \in H$ such that $z \perp F$. Hence, $F_1 = F \cup \{e\}$, where $e = \|z\|^{-1}z$ is orthonormal, and F is a proper subset of F_1. This contradicts the maximality of F. ■

2 Hahn-Banach Theorem

2.1 What is the Hahn-Banach Theorem?

- It is an extension theorem for linear functionals in normed spaces (in a real vector space).
- It guarantees the abundance of bounded linear functionals on a normed space.
- It characterizes the extent to which values of a linear functional can be pre-assigned.

Roughly speaking, when we talk about extending an object, we usually refer to preserving desired properties from one space to another. More specifically, the object of interest in the Hahn-Banach theorem is a linear functional f that is defined on a subspace Z of a vector space X and satisfies a certain boundedness property which will be represented in terms of a **sublinear functional**.

Definition (Sublinear Functional). We say that p defined on a vector space X is a **sublinear functional** if

(1) $p(x + y) \leq p(x) + p(y)$ for all $x, y \in X$; that is, p is **subadditive**

(2) $p(\alpha x) = \alpha p(x)$ for all $\alpha \geq 0$ in $\mathbb{R}$ and $x \in X$; that is, p is **positive-homogeneous**.

- We assume that the functional f to be extended is majorized (to be bounded) by a functional p (that is defined on X) that satisfies the above properties.
- We will extend f to a functional $\tilde{f}$ which will retain the boundedness properties that f has on X instead of the subset of X.
- This version of the theorem will assume that X will be a real vector space.

Theorem (Hahn-Banach Theorem (Extension of linear functionals)). Let X be real vector space and p a sublinear functional on X. Furthermore, let f be a linear functional which is defined on a subspace Z of X and satisfies

$$f(x) \leq p(x) \quad \forall x \in Z.$$

Then f has a linear extension $\tilde{f}$ from Z to X satisfying

$$\tilde{f}(x) \leq p(x) \quad \forall x \in X; \tag{*}$$

that is, $\tilde{f}$ is a linear functional on X, satisfies (*) on X and $\tilde{f}(x) = f(x)$ for every $x \in Z$.

Proof. We will proceed using the following steps below:

(a) The set E of all linear extensions of g of f satisfying $g(x) \leq p(x)$ on their domain $D(g)$ can be partially ordered and Zorn's lemma yields a maximal element of $\tilde{f}$ of E.

(b) $\tilde{f}$ is defined on the entire space X.

(c) An auxiliary relation which was used in (b).

To start, we will prove (a). Let E be the set of all linear extensions g of f for which

$$g(x) \leq p(x) \quad \forall x \in D(g).$$

Note that $E \neq \emptyset$ since $f \in E$ by assumption. On E we can define a partial ordering by $g \leq h$ meaning h is an extension of g, that is, by definition, $D(h) \supseteq D(g)$ and $h(x) = g(x)$ for every $x \in D(g)$. For any chain $C \subseteq E$, we now define $\hat{g}$ by

$$\hat{g}(x) = g(x) \quad \text{if } x \in D(g) \qquad (g \in C)$$

where it can be proven relatively easily that $\hat{g}$ is a linear functional with the domain being

$$D(\hat{g}) = \bigcup_{g \in C} D(g)$$

which is a vector space since C is a chain. The definition of $\hat{g}$ is well-defined. Indeed, for $x \in D(g_1) \cap D(g_2)$ with $g_1, g_2 \in C$, we have $g_1(x) = g_2(x)$ since C is a chain so that $g_1 \leq g_2$ or $g_2 \leq g_1$. Clearly, $g \leq \hat{g}$ for all $g \in C$. Hence, $\hat{g}$ is an upper bound of C. Since $C \subseteq E$ was arbitrary, it follows from Zorn's lemma that E contains a maximal element $\tilde{f}$. By the definition of E, this is a linear extension of f which satisfies

$$\tilde{f}(x) \leq p(x) \quad . \qquad (x \in D(\tilde{f}))$$

Now, we will show that $D(\tilde{f})$ is all of X. Suppose for contradiction that $D(\tilde{f}) \neq X$. Then we can choose a $y_1 \in X \setminus D(\tilde{f})$ and consider the subspace $Y_1 = \operatorname{span}(D(\tilde{f}) \cap \{y_1\})$. Note that $y_1 \neq 0$ since $0 \in D(\tilde{f})$. Any $x \in Y_1$ can be written as

$$x = y + \alpha y_1.. \qquad (y \in D(\tilde{f}))$$

Note that this representation is unique. Indeed, $y + \alpha y_1 = \tilde{y} + \beta y_1$ with $\tilde{y} \in D(\tilde{f})$ implies that

$$y - \tilde{y} = (\beta - \alpha) y_1.$$

Since $y_1 \notin D(\tilde{f})$, the only solution to the equation above is for $y - \tilde{y} = 0$ and $\beta - \alpha = 0$. This tells us now that our representation is unique.

A functional g_1 on Y_1 is defined by

$$g_1(y + \alpha y_1) = \tilde{f}(y) + \alpha c$$

where c is any real constant. It is not difficult to see that g_1 is linear. Furthermore, for $\alpha = 0$, we have $g_1(y) = \tilde{f}(y)$. Hence, g_1 is a proper extension of $\tilde{f}$, that is, an extension such that $D(\tilde{f})$ is a proper subset of $D(g_1)$. Thus, proving that $g_1 \in E$ by showing that $g_1(x) \leq p(x)$ for all $x \in D(g_1)$ will contradict the maximality of $\tilde{f}$ and so the fact that $D(\tilde{f}) = X$ must be true.

Indeed, we must show that this is the case for a suitable c that satisfies the above desired result. We consider any y and z in $D(\tilde{f})$. We have

$$\begin{aligned} \tilde{f}(y) - \tilde{f}(z) = \tilde{f}(y - z) &\leq p(y - z) \\ &= p(y + y_1 - y_1 - z) \\ &\leq p(y + y_1) + p(-y_1 - z). \end{aligned}$$

Taking the last term to the left and the term $\tilde{f}(y)$ to the right, we have

$$-p(-y_1 - z) - \tilde{f}(z) \le p(y + y_1) - \tilde{f}(y),$$

where y_1 is fixed.

■

3 Hahn-Banach Theorem for Complex Vector Spaces and Normed Spaces

Theorem (Generalized Hahn-Banach Theorem). Let X be a real or complex vector space and p a real-valued functional on X which is subadditive, that is, for all $x, y \in X$,

$$p(x + y) \le p(x) + p(y), \tag{1}$$

and for every scalar α satisfies

$$p(\alpha x) = |\alpha| p(x).$$

Furthermore, let f be a linear functional which is defined on a subspace Z of X and satisfies

$$|f(x)| \le p(x) \quad \forall x \in Z.$$

Then f has a linear extension $\tilde{f}$ from Z to X satisfying

$$|\tilde{f}(x)| \le p(x). \quad \forall x \in X$$

Proof. ■

4 Adjoint Operator

4.1 The Adjoint Operator

- From the Hahn-Banach Theorem, we can see that for every bounded linear operator $T : X \to Y$ on a normed space X, we can make sense of and associate an adjoint operator $T^\times$ of T.
- This helps us solve equations in physics and other applications.
- For this section, we will look at some of its properties and how it is related to the Hilbert-adjoint operator T^* defined in earlier sections of this book. **Note:** When we refer to the Adjoint Operator of T, we are not referring to the Hilbert-adjoint Operator of T. These two concepts are distinct from each other.
- To this end, let us first consider a bounded linear operator $T : X \to Y$, where X and Y are normed spaces.

Let g be any bounded linear functional on Y. Since g is defined on all of Y, we can set $y = Tx$ and obtain a functional f such that

$$f(x) = g(Tx) \quad \forall x \in X. \tag{1}$$

- Notice that f is linear since g and T are linear.
- f is bounded since g and T are both bounded. That is,

  $$|f(x)| = |g(Tx)| \le \|g\| \|Tx\| \le \|g\| \|T\| \|x\|.$$

  Taking the supremum over all $x \in X$ where $\|x\| = 1$, we obtain

  $$\|f\| \le \|g\| \|T\|. \tag{2}$$

  Thus, (2) tells us that $f \in X'$, where X' is the dual space of X.

- (1) tells us that f is an operator defined from Y' into X'. This is precisely the **adjoint operator** of T which will be defined more explicitly below.

Definition (Adjoint Operator $T^\times$). Let $T : X \to Y$ be a bounded linear operator, where X and Y are normed spaces. Then the **adjoint operator** $T^\times : Y' \to X'$ of T is defined by

$$f(x) = (T^\times g)(x) = g(Tx) \quad \forall g \in Y'.$$

We will see in the next theorem that the adjoint operator $T^\times$ of T will have the same norm as the operator itself. To do this, we will need Theorem 4.3-3 (a result that followed from the Hahn-Banach Theorem) to establish this property.

4.2 Equivalent Norms

Theorem (Norm Of The Adjoint Operator (4.5-2)). The adjoint operator $T^\times$ is linear and bounded, and

$$\|T^\times\| = \|T\|. \tag{5}$$

Proof. The operator $T^\times$ is linear since its domain Y' is a vector space. Thus, we obtain

$$\begin{aligned}(T^\times(\alpha g_1 + \beta g_2))(x) &= (\alpha g_1 + \beta g_2)(Tx)\\ &= \alpha g_1(Tx) + \beta g_2(Tx)\\ &= \alpha(T^\times g_1)(x) + \beta(T^\times g_2)(x).\end{aligned}$$

We will prove (5) from (4). That is, we will show that $\|T^\times\| \leq \|T\|$ and $\|T^\times\| \geq \|T\|$. For $f = T^\times g$, we can use (2) to write

$$\|T^\times g\| = \|f\| \leq \|g\|\|T\|.$$

Taking the supremum over all $g \in Y'$ of norm one, we obtain

$$\|T^\times\| \leq \|T\|$$

proving the first inequality. To get the other inequality, we will use Theorem 4.3-3. That is, for every $x_0 \neq 0$ in X, there exists a $g_0 \in Y'$ such that

$$\|g_0\| = 1 \quad \text{and} \quad g_0(Tx_0) = \|Tx_0\|.$$

Notice, by definition, that $g_0(Tx_0) = (T^\times g_0)(x_0)$. Setting $f_0 = T^\times g_0$, we obtain

$$\begin{aligned}\|Tx_0\| = g_0(Tx_0) &= f_0(x_0)\\ &\leq \|f_0(x_0)\|\\ &\leq \|f_0\|\|x_0\|\\ &= \|T^\times g_0\|\|x_0\|\\ &\leq \|T^\times\|\|g_0\|\|x_0\|.\end{aligned}$$

Since $\|g_0\| = 1$, we further have

$$\|Tx_0\| \leq \|T^\times\|\|x_0\|$$

which holds for $x_0 = 0$ since $T0 = 0$. Also, we have

$$\|Tx_0\| \leq \|T\|\|x_0\|$$

where $\|T\|$ is the least upper bound by definition and $\|T^\times\|$ is an upper bound of $\|Tx_0\|$. Hence, $\|T^\times\| \geq \|T\|$ giving us the second inequality. Thus, we conclude that $\|T^\times\| = \|T\|$. ■

Example (Adjoint of a Linear Operator in $\mathbb{R}^n$). Given a linear operator $T : \mathbb{R}^n \to \mathbb{R}^n$, the matrix representation $T_E = (\tau_{jk})$, its adjoint is just the transpose of T_E where $E = \{e_1, \ldots, e_n\}$ is a basis for $\mathbb{R}^n$ whose elements are arranged in an order that is fixed. Let $x = (\xi_1, \ldots, \xi_n)$ and $y = (\eta_1, \ldots, \eta_n)$ be two column vectors in $\mathbb{R}^n$. Let $F = \{f_1, \ldots, f_n\}$ be the dual basis for the dual of $\mathbb{R}^n$. Given that every linear functional g in the dual of $\mathbb{R}^n$ has the representation,

$$g = \alpha_1 f_1 + \cdots + \alpha_n f_n,$$

the formula for such an adjoint is given by

$$T_E^{\times} g(x) = g(T_E x) = \sum_{k=1}^{n} \beta_k \xi_k \quad \text{where} \quad \beta_k = \sum_{k=1}^{n} \tau_{jk} \alpha_j.$$

This tells us that for every linear operator T from $\mathbb{R}^n$ to $\mathbb{R}^n$ with matrix representation T_E, the adjoint operator $T^{\times}$ is the transpose of T_E. The same result holds if T is a linear operator from $\mathbb{C}^n$ to $\mathbb{C}^n$.

Below are some properties of the adjoint operator:

- For every $S, T \in B(X, Y)$, we have $(S + T)^{\times} = S^{\times} + T^{\times}$.
- $(\alpha T)^{\times} = \alpha T^{\times}$.
- Let X, Y, Z be normed spaces and $T \in B(X, Y)$ and $S \in B(Y, Z)$. Then the adjoint operator of ST is
$$(ST)^{\times} = T^{\times} S^{\times}.$$
- $(T^{\times})^{-1} = (T^{-1})^{\times}$.

4.3 Relation Between Adjoint Operator and Hilbert-Adjoint Operator

If X and Y are both Hilbert spaces, then we can express the Hilbert-adjoint operator T^* of a linear operator T on a Hilbert space in terms of the adjoint operator $T^{\times}$ of T.

Let $X = H_1$ and $Y = H_2$ where H_1 and H_2 are two Hilbert spaces. Consider $T : H_1 \to H_2$ and $T_{\times} : H_2' \to H_1'$ and recall that the adjoint $T^{\times}$ is defined by

$$T^{\times} g = f \tag{a}$$
$$g(Tx) = f(x) \tag{b}$$

where $f \in H_1'$ and $g \in H_2'$. Now that we have two Hilbert spaces H_1 and H_2 to work with, we can express each linear functional f and g from H_1 and H_2, respectively, via the Riesz representation; that is, we can find $x_0 \in H_1$ and $y_0 \in H_2$ such that

$$f(x) = \langle x, x_0 \rangle$$
$$g(y) = \langle y, y_0 \rangle.$$

Furthermore, we see that x_0 and y_0 are uniquely determined by f and g (using Theorem 3.8-1), respectively, thereby giving us the following operators $A_1 : H_1' \to H_1$ and $A_2 : H_2' \to H_2$ defined by $A_1 f = x_0$ and $A_2 g = y_0$. From the same theorem, A_1 and A_2 are both:

(a) bijections;

(b) isometries because $\|A_1 f\| = \|x_0\| = \|f\|$ and $\|A_2 g\| = \|y_0\| = \|g\|$;

(c) conjugate linear.

Composing $A_2^{-1}, T^{\times}$, and A_1 together, we define

$$A_1 T^{\times} A_2^{-1} : H_2 \to H_1 \quad \text{by} \quad A_1 T^{\times} A_2^{-1} y_0 = x_0.$$

Our claim is that $T^* = A_1 T^{\times} A_2^{-1}$. Since we are able to take advantage of the Riesz representations of each linear functional from their respective duals, it follows that

$$\langle Tx, y_0 \rangle = g(Tx) = f(x) = \langle x, x_0 \rangle = \langle x, (A_1 T^{\times} A_2) y_0 \rangle.$$

Since $\langle Tx, y_0 \rangle = \langle x, T^* y_0 \rangle$, we conclude that $A_1 T^{\times} A_2^{-1} = T^*$.

4.4 Key Differences Between Adjoints and Hilbert-Adjoint

- The Hilbert-adjoint T^* is defined directly on the spaces X and Y whereas the adjoint $T^\times$ is defined on the duals of X and Y, respectively.
- For $\alpha \in \mathbb{F}$, we have $(\alpha T)^\times = \alpha T^\times$ whereas $(\alpha T^*) = \overline{\alpha} T^*$.
- For finite dimensional Euclidean spaces, the adjoint $T^\times$ represents the transpose the matrix representation of T and the Hilbert-adjoint T^* represents the conjugate transpose of the same matrix.

5 Reflexive Spaces

5.1 Review

- Recall that the dual space of a normed space X is the set of all continuous linear functionals that map elements from the normed space into some scalar field $\mathbb{F}$. That is,

$$X' = \{f : X \to \mathbb{F} : f \text{ is a continuous linear functional}\}$$

- The double dual on the other hand is the set of all continuous linear functionals that map elements of the dual (which are itself the set of all continuous linear functionals) to the same scalar field $\mathbb{F}$. In set notation, we have

$$X'' = \{F : X' \to \mathbb{F} : F \text{ is a continuous linear functional}\}.$$

- There is a natural way of relating a normed space with its double dual . This is done through the **canonical mapping** $C : X \to X''$ which is defined by $x \mapsto g_x$ where $g_x \in X''$. That is,

$$g_x(f) = f(x) \quad f \in X'. \tag{1}$$

  Keep in mind here that the element $x \in X$ is fixed and gets evaluated by some element f in the dual space X'.

- If f is bounded, then g_x above will also be bounded.

5.2 Canonical Mapping

Lemma (Norm of g_x)**.** For every fixed x in a normed space X, the functional g_x defined by (1) is a bounded linear functional on X', so that $g_x \in X''$, and has the norm

$$\|g_x\| = \|x\|.$$

Proof. First, we show that g_x is linear. For all $f_1, f_2 \in X'$ and for all $\alpha \in \mathbb{F}$, we have

$$\begin{aligned} g_x(\alpha f_1 + f_2) &= (\alpha f_1 + f_2)(x) \\ &= (\alpha f_1)(x) + f_2(x) \\ &= \alpha f_1(x) + f_2(x) \\ &= \alpha g_x(f_1) + g_x(f_2). \end{aligned}$$

Using Corollary 4.3-4, we get that

$$\|g_x\| = \sup_{\substack{f \in X' \\ f \neq 0}} \frac{|g_x(f)|}{\|f\|} = \sup_{\substack{f \in X' \\ f \neq 0}} \frac{|f(x)|}{\|f\|} = \|x\|.$$

■

Definition (Canonical Mapping). For every $x \in X$, there exists a unique bounded linear functional $g_x \in X''$ given by the mapping $C : X \to X''$ by $x \mapsto g_x$ which is called the **canonical mapping** of X into X''.

Lemma (Canonical Mapping). The canonical mapping C given above is an isomorphism of the normed space X onto the normed space $R(C)$, the range of C.

Proof. First, we prove that C is linear. Let $\alpha, \beta \in \mathbb{F}$ and $x, y \in X$. Then

$$g_{\alpha x+\beta y}(f) = f(\alpha x + \beta y) = \alpha f(x) + \beta f(y) = \alpha g_x(f) + \beta g_y(f).$$

In particular, we see that $g_x - g_y = g_{x-y}$. Indeed, for all $f \in X'$, we have

$$\begin{aligned}(g_x - g_y)(f) &= g_x(f) - g_y(f) \\ &= f(x) - f(y) \\ &= f(x - y) && (f \text{ is linear}) \\ &= g_{x-y}(f).\end{aligned}$$

Hence, from the lemma at the beginning of this section, we have

$$\|g_x - g_y\| = \|g_{x-y}\| = \|x - y\|$$

which proves that C is an isometry and that C is injective as a result. Hence, C is bijective (surjectivity is readily seen from the above equality). ■

Definition (Embeddable). We say that a normed space X is **embeddable** in a normed space Z if X is isomorphic with a subspace of Z. (This is an isomorphism of normed spaces that preserves the norm defined on X)

Note that C is not always onto; that is, its range in general is a proper subspace of X''. When C is onto, we say that X is **reflexive**.

Definition (Reflexivity). A normed space X is said to be **reflexive** if

$$R(C) = X''$$

where $C : X \to X''$ is the canonical mapping that was defined previously.

- Given a reflexive space X, we have $X \cong X''$ by lemma 4.6-2.
- Note that the converse of the above does not necessarily hold.

Theorem (Completeness). If a normed space X is reflexive, it is complete (hence a Banach space).

Proof. Since X'' is the dual space of X', it is complete by Theorem 2.10-4. Reflexivity of X implies that $R(C) = X''$. Completeness of X now follows that of lemma 4.6-2. Indeed, let (x_n) be a Cauchy sequence in X. First, we will prove that g_{x_n} is also a Cauchy sequence in X''. Indeed, we see this from lemma 4.6-2; that is,

$$\|g_{x_n} - g_{x_m}\| = \|g_{x_n - x_m}\| = \|x_n - x_m\| \longrightarrow 0 \text{ as } n, m \longrightarrow \infty.$$

Since X'' is complete, there exists a $g_x \in X''$ (via the canonical map $x \mapsto g_x$) such that $g_{x_n} \to g_x$. That is,

$$\|x_n - x\| = \|g_{x_n - x}\| = \|g_{x_n} - g_x\| \longrightarrow 0 \text{ as } n, m \longrightarrow \infty.$$

■

It follows from the fact that X is a finite dimensional space that X will be reflexive. An immediate example of this is $\mathbb{R}^n$.

Theorem (Finite Dimension). Every finite dimensional normed space is reflexive.

Theorem (Hilbert Space). Every Hilbert space H is reflexive.

Proof. We will show that the canonical map $C : H \to H''$ is surjective, by showing that for every $g \in H''$, there exists an $x \in H$ such that $Cx = g$.

Define $A : H' \to H$ by $Af = z$, where z is given by the Riesz representation $f(x) = \langle x, z \rangle$ by Theorem 3.8-1. Note that z depends on f and is uniquely determined by f and has norm $\|z\| = \|f\|$. We see that A is bijective, isometric, and conjugate linear. Note that H' has the inner product $\langle f_1, f_2 \rangle_1 = \langle Af_2, Af_1 \rangle$ verified back in section 3.1.

Now, let's begin the proof by letting $g \in H''$ be arbitrary. Using the Riesz's representation, we get that

$$g(f) = \langle f, f_0 \rangle_1 = \langle Af_0, Af \rangle. \tag{*}$$

Earlier we said that $f(x) = \langle x, z \rangle$ where $z = Af$. In particular, set $Af_0 = x$, and so we have

$$\langle Af_0, Af \rangle = \langle x, z \rangle = f(z). \tag{**}$$

Now, (*) and (**) implies that $g(f) = f(x)$ where $g = Cx$ by definition of the Canonical map. Thus, C must be surjective (that is, $R(C) = H''$) and so H is reflexive ■

Lemma (Existence of a Functional). Let Y be a proper closed subspace of a normed space X. Let $x_0 \in X - Y$ be arbitrary and

$$\delta = \inf_{\tilde{y} \in Y} \|\tilde{y} - x_0\|$$

the distance from x_0 to Y. Then there exists an $\tilde{f} \in X'$ such that

$$\|\tilde{f}\| = 1 \quad \tilde{f}(y) = 0 \ \forall y \in Y \quad \tilde{f}(x_0) = \delta.$$

Proof. Consider the subspace $Z = \text{span}(Y \cup \{x_0\})$ and define a bounded linear functional f on Z by

$$f(z) = f(y + \alpha x_0) = \alpha\delta$$

where $z = y + \alpha x_0$ is the unique representation of every element z in Z. First, we show that f is indeed linear. Let $u, v \in Z$ and $t \in \mathbb{F}$. Then we have

$$u = y_1 + \alpha_1 x_0 \text{ and } v = y_2 + \alpha_2 x_0$$

for some $y_1, y_2 \in Y$ and $\alpha_1, \alpha_2 \in \mathbb{F}$. Then we have

$$\begin{aligned} f(tu+v) &= f(t(y_1+\alpha_1 x_0)+(y_2+\alpha_2 x_0)) \\ &= f((ty_1+y_2)+(t\alpha_1+\alpha_2)x_0) \\ &= (t\alpha_1+\alpha_2)\delta \\ &= t\alpha_1\delta+\alpha_2\delta \\ &= tf(u)+f(v). \end{aligned}$$

Also, since Y is closed, we have $\delta > 0$ so that $f \neq 0$. Now, $\alpha = 0$ gives $f(y) = 0$ for all $y \in Y$. If $\alpha = 1$ and $y = 0$, we have $f(x_0) = \delta$. Now, we will show that f is bounded on Z. Indeed, we will show that $\|f\| = 1$. Since Y is a subspace, we know that $-(1/\alpha)y \in Y$. We obtain from this the following result:

$$\begin{aligned} |f(z)| = |\alpha\delta| &= |\alpha|\delta \\ &= |\alpha|\Big\| -\frac{1}{\alpha}y - x_0\Big\| \\ &= \|y+\alpha x_0\| \\ &= \|z\|. \end{aligned}$$

Hence, $|f(z)| \leq \|z\|$ and so

$$\frac{|f(z)|}{\|z\|} \leq 1 \implies \|f\| \leq 1$$

where $\|f\| = \sup_{\|z\|\neq 0} \frac{|f(z)|}{\|z\|}$. Now, we need to show that $\|f\| \geq 1$. Indeed, using the definition of the infimum, we can find a sequence (y_n) such that $\|y_n - x_0\| \to \delta$. Define $z_n = y_n - x_0$. With this representation, we find $f(z_n) = -\delta$ with $\alpha = -1$. Hence,

$$\|f\| = \sup_{\substack{z\in Z \\ z\neq 0}} \frac{|f(z)|}{\|z\|} \geq \frac{|f(z_n)|}{\|z_n\|} = \frac{\delta}{\|z_n\|} = \frac{\delta}{\|z_n\|} \longrightarrow \frac{\delta}{\delta} = 1$$

since $\|z_n\| \to \delta$. Thus, $\|f\| \geq 1$ so that $\|f\| = 1$. Using the Hahn-Banach theorem for normed spaces, it follows that we can extend f to X without changing the norm. ■

Theorem (Separability). If the dual space X' of a normed space X is separable, then X itself is separable.

Proof. Suppose X' is separable. Then the unit sphere $U' = \{f : \|f\| = 1\} \subseteq X'$ also contains a countable dense subset (f_n). Since $f_n \in U'$, it follows that

$$\|f_n\| = \sup_{\|x\|=1} |f_n(x)| = 1.$$

Using the definition of the supremum, there exists a sequence (x_n) in X with norm 1 such that $|f_n(x)| \geq \frac{1}{2}$.

Let $Y = \overline{\operatorname{span}(x_n)}$. Note that Y is separable because Y contains a dense subset, namely, the set of all linear combinations of the $x_n's$ with coefficients whose real and imaginary parts are rational.

We need to show that $Y = X$. Suppose for contradiction that $Y \neq X$. Then since Y is closed, we know (by Lemma 4.6-7) that there exists an $\tilde{f} \in X'$ with $\|\tilde{f}\| = 1$ and $\tilde{f}(y) = 0$ for all $y \in Y$.

Since $x_n \in Y$ for all $n \in \mathbb{N}$, we have $\tilde{f}(x_n) = 0$ and so

$$\begin{aligned}\frac{1}{2} \leq |f_n(x_n)| &= |f_n(x_n) - \tilde{f}(x_n)| \\ &= |(f_n - \tilde{f})(x_n)| \\ &= \|f_n - \tilde{f}\|\|x_n\|\end{aligned}$$

where $\|x_n\| = 1$. Thus, $\|f_n - f\| \geq \frac{1}{2}$, but note that we assume that (f_n) is dense in U' where $\tilde{f}$ is in U' which is a contradiction. Hence, $Y = X$ and we are done. ■

6 Uniform Boundedness Theorem

Definition (Category). A subset M of a metric space X is said to be

(a) **rare** (or **nowhere dense**) in X if its closure $\overline{M}$ has no interior points,

(b) **meager** (or **of the first category**) in X if M is the union of countably many sets each of which is rare in X,

(c) **nonmeager** (or of the **second category**) in X if M is not meager in X.

Theorem (Baire's Category Theorem (Complete Metric Spaces)). If a metric space $X \neq \emptyset$ is complete, it is nonmeager in itself. Hence, if $X \neq \emptyset$ is complete and

$$X = \bigcup_{k=1}^{\infty} A_k$$

where A_k is closed, then at least one A_k contains a nonempty open subset.

Theorem (Uniform Boundedness Theorem). Let (T_n) be a sequence of bounded linear operators $T_n : X \to Y$ from a Banach Space X into a normed space Y such that $(\|T_n x\|)$ is bounded for every $x \in X$, say,

$$\|T_n x\| \leq c_x \quad \forall n \in \mathbb{N},$$

where c_x is a real number. Then the sequence of the norms $\|T_n\|$ is bounded, that is, there is a c such that

$$\|T_n\| \leq c.$$

7 Strong and Weak Convergence

Definition (Strong Convergence). A sequence (x_n) in a normed space X is said to be **strongly convergent** (or **convergent in the norm**) if there is an $x \in X$ such that

$$\lim_{n\to\infty} \|x_n - x\| = 0.$$

Notationally, this is written as $\lim\limits_{n\to\infty} x_n = x$ or simply $x_n \to x$ where x is called the **strong limit** of (x_n) and, we say that (x_n) **strongly to** x.

Definition (Weak Convergence). A sequence (x_n) in a normed space X is said to be **weakly con-**

vergent if there is an $x \in X$ such that for every $f \in X'$,

$$\lim_{n\to\infty} f(x_n) = f(x)$$

where notationally, we say $x_n \xrightarrow{w} x$ and we call x as the **weak limit** of (x_n) and we say that (x_n) **converges weakly to** x.

Lemma (Weak Convergence). Let (x_n) be a **weakly convergent sequence** in a normed space X, say $x_n \xrightarrow{w} x$. Then:

(a) The weak limit of (x_n) is unique.

(b) Every subsequence of (x_n) converges weakly to x.

(c) The sequence $(\|x_n\|)$ is bounded.

Theorem (Strong and Weak Convergence). Let (x_n) be a sequence in a normed space X. Then:

(a) Strong convergence implies weak convergence with the same limit.

(b) The converse of (a) is not generally true.

(c) If $\dim(X) < \infty$, then weak convergence implies strong convergence.

Lemma (Weak Convergence). In a normed space X, we have $x_n \xrightarrow{w}$ if and only if

(A) The sequence $(\|x_n\|)$ is bounded.

(B) For every $f \in M \subseteq X'$, we have $f(x_n) to f(x)$.

8 Convergence of Sequences of Operators and Functionals

Definition (Convergences of Sequences of Operators). Let X and Y be normed spaces. A sequence (T_n) of operators $T_n \in B(X, Y)$ is said to be:

(1) **Uniformly Operator Convergent** if (T_n) converges in the norm on $B(X, Y)$

(2) **Strongly Operator Convergent** if $(T_n x)$ converges strongly in Y for every $x \in X$.

(3) **Weakly Operator Convergent** if $(T_n x)$ converges weakly in Y for every $x \in X$.

The above tells us that there exists an operator $T : X \to Y$ such that

(1) $\|T_n - T\| \longrightarrow 0$ (Uniform)

(2) $\|T_n x - Tx\| \longrightarrow 0$ for all $x \in X$ (Strong Operator Convergence)

(3) $|f(T_n x) - f(Tx)| \longrightarrow 0$ for all $x \in X$ and all $f \in Y'$ (Weak Operator Convergence)

Definition (Strong and Weak* Convergence of a Sequence of Funcionals). Let (f_n) be a sequence of bounded linear functionals on a normed space X. Then:

(a) **Strong Convergence** of (f_n) means that there is an $f \in X'$ such that $\|f_n - f\| \longrightarrow 0$. This is written

$$f_n \longrightarrow f.$$

(b) **Weak* Convergence** of (f_n) means that there is an $f \in X'$ such that $f_n(x) \longrightarrow f(x)$ for all $x \in X$. This is written

$$f_n \xrightarrow{w*} f.$$

9 Open Mapping Theorem

Definition (Open Mapping). Let X and Y be metric spaces. Then $T : D(T) \to Y$ with domain $D(T) \subseteq X$ is called an **open mapping** if for every open set in $D(T)$ the image is an open set in Y.

Lemma (Open Unit Ball). A bounded linear operator T from a Banach space X onto a Banach space Y has the property that the image $T(B_0)$ of the open ball $B_0 = B(0; 1) \subseteq X$ contains an open ball about $0 \in Y$.

Theorem (Open Mapping Theorem/Bounded Inverse Theorem). A bounded linear operator T from a Banach space X onto a Banach space Y is an open mapping. Hence, if T is bijective, T^{-1} is continuous and thus bounded.

10 Closed Linear Operators/Closed Graph Theorem

Definition (Closed Linear Operator). Let X and Y be normed spaces and $T : D(T) \to Y$ a linear operator with domain $D(T) \subseteq X$. Then T is called a **linear operator** if its graph

$$\mathcal{G}(T) = \{(x, y) : x \in D(T), y = Tx\}$$

is closed in the normed space $X \times Y$, where the two algebraic operations of a vector space in $X \times Y$ are defined as usual, that is,

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$
$$\alpha(x, y) = (\alpha x, \alpha y)$$

and the norm on $X \times Y$ is defined by

$$\|(x, y)\| = \|x\| + \|y\|.$$

Theorem (Closed Graph Theorem). Let X and Y be Banach spaces and $T : D(T) \to Y$ a closed linear operator, where $D(T) \subseteq X$. Then if $D(T)$ is closed in X, the operator T is bounded.

Theorem (Closed Linear Operator). Let $T : D(T) \to Y$ be a linear operator, where $D(T) \subseteq X$ and X and Y are normed spaces. Then T is closed if and only if it has the following property. If $x_n \to x$, where $x_n \in D(T)$, and $Tx_n \to y$, then $x \in D(T)$ and $Tx = y$.

Lemma (Closed Operator). Let $T : D(T) \to Y$ be a bounded linear operator with domain $D(T) \subseteq X$, where X and Y are normed spaces. Then:

(a) If $D(T)$ is a closed subset of X, then T is closed.

(b) If T is closed and Y is complete, then $D(T)$ is closed subset of X